

31ST.py - C:/Users/thala/Downloads/FODS/31ST.py (3.14.3)

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```
import pandas as pd
from sklearn.cluster import KMeans
```

```
# Sample customer data
```

```
data = {
    "Purchase": [1000, 1200, 1500, 5000, 5500, 6000],
    "Visits": [5, 6, 7, 20, 22, 25]
}
```

```
df = pd.DataFrame(data)
```

```
# Select features
```

```
X = df[["Purchase", "Visits"]]
```

```
# Create K-Means model
```

```
model = KMeans(n_clusters=2, random_state=42, n_init=10)
df["Segment"] = model.fit_predict(X)
```

```
print(df)
```

IDLE Shell 3.14.3

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Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:\Users\thala\Downloads\FODS\31ST.py =====

	Purchase	Visits	Segment
0	1000	5	0
1	1200	6	0
2	1500	7	0
3	5000	20	1
4	5500	22	1
5	6000	25	1

>>>

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```

32ND.py - C:/Users/thala/Downloads/FODS/32ND.py (3.14.3)
File Edit Format Run Options Window Help
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# House size and price data
size = np.array([1000, 1200, 1500, 1800, 2000]).reshape(-1, 1)
price = np.array([50, 60, 75, 90, 100])

# Create model
model = LinearRegression()
model.fit(size, price)

# Prediction
predicted_price = model.predict(size)

# Evaluation
print("MSE:", round(mean_squared_error(price, predicted_price), 2))
print("R2 Score:", round(r2_score(price, predicted_price), 2))

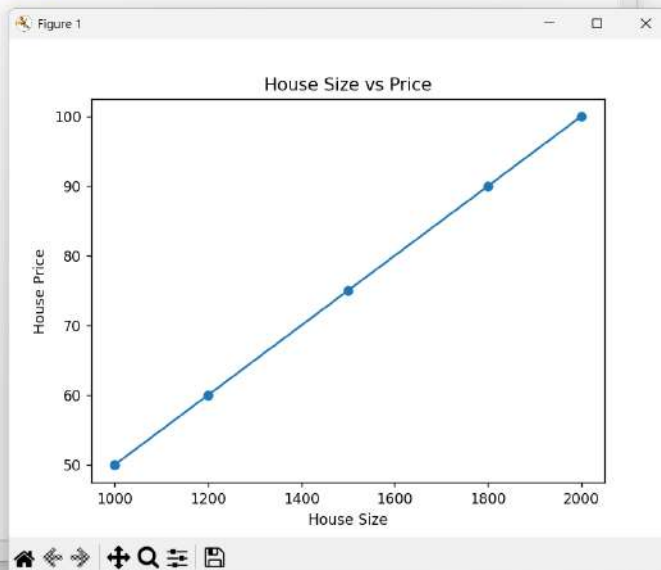
# Plot
plt.scatter(size, price)
plt.plot(size, predicted_price)
plt.xlabel("House Size")
plt.ylabel("House Price")
plt.title("House Size vs Price")
plt.show()

```

```

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AMD64)] on win32
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>>>
===== RESTART: C:/Users/thala/Downloads/FODS/32ND.py =====
MSE: 0.0
R2 Score: 1.0

```



```
33RD.py - C:/Users/thala/Downloads/FODS/33RD.py (3.14.3)
File Edit Format Run Options Window Help
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Sample car data
data = {
    "EngineSize": [1.2, 1.5, 1.8, 2.0, 2.5],
    "HorsePower": [80, 100, 120, 150, 180],
    "Price": [500000, 700000, 900000, 1200000, 1500000]
}

df = pd.DataFrame(data)

# Features and target
X = df[["EngineSize", "HorsePower"]]
y = df["Price"]

# Train model
model = LinearRegression()
model.fit(X, y)

# Prediction
predicted = model.predict(X)

# Evaluation
print("MSE:", round(mean_squared_error(y, predicted), 2))
print("R2 Score:", round(r2_score(y, predicted), 2))

print("\nCoefficients:")
print("Engine Size:", round(model.coef_[0], 2))
print("HorsePower:", round(model.coef_[1], 2))
```

```
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>>>
===== RESTART: C:/Users/thala/Downloads/FODS/33RD.py =====
MSE: 0.0
R2 Score: 1.0

Coefficients:
Engine Size: -0.0
HorsePower: 10000.0

>>>
```

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```
34TH.py - C:/Users/thala/Downloads/FODS/34TH.py (3.14.3)
File Edit Format Run Options Window Help
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
```

```
# Sample data: [Age, Blood Pressure, Cholesterol]
```

```
X = [
    [25, 120, 180],
    [30, 125, 190],
    [35, 130, 200],
    [40, 140, 220],
    [45, 150, 240],
    [50, 160, 260],
    [55, 170, 280],
    [60, 180, 300]
]
```

```
# Good = 1, Bad = 0
y = [1, 1, 1, 1, 0, 0, 0, 0]
```

```
# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)
```

```
# Train KNN model
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train, y_train)
```

```
# Prediction
y_pred = model.predict(X_test)
```

```
# Evaluation
print("Predictions:", y_pred)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred, zero_division=0))
print("Recall:", recall_score(y_test, y_pred, zero_division=0))
print("F1 Score:", f1_score(y_test, y_pred, zero_division=0))
```

```
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>>>
===== RESTART: C:/Users/thala/Downloads/FODS/33RD.py =====
MSE: 0.0
R2 Score: 1.0

Coefficients:
Engine Size: -0.0
HorsePower: 10000.0
>>>
===== RESTART: C:/Users/thala/Downloads/FODS/34TH.py =====
Predictions: [1 0]
Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1 Score: 1.0
>>>
|
```

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```

35TH.py - C:/Users/thala/Downloads/FODS/35TH.py (3.14.3)
File Edit Format Run Options Window Help

import pandas as pd
from sklearn.cluster import KMeans

# Transaction data
data = {
    "CustomerID": [1, 2, 3, 4, 5, 6],
    "AmountSpent": [500, 700, 800, 5000, 6000, 7000],
    "VisitFrequency": [2, 3, 4, 15, 18, 20]
}

df = pd.DataFrame(data)

# Select features
X = df[["AmountSpent", "VisitFrequency"]]

# K-Means clustering
model = KMeans(n_clusters=2, random_state=42, n_init=10)
df["Segment"] = model.fit_predict(X)

# Display result
print(df)

```

```

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>>>
===== RESTART: C:/Users/thala/Downloads/FODS/33RD.py =====
MSR: 0.0
R2 Score: 1.0

Coefficients:
Engine Size: -0.0
HorsePower: 10000.0

>>>
===== RESTART: C:/Users/thala/Downloads/FODS/34TH.py =====
Predictions: [1 0]
Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1 Score: 1.0

>>>
===== RESTART: C:/Users/thala/Downloads/FODS/35TH.py =====

```

	CustomerID	AmountSpent	VisitFrequency	Segment
0	1	500	2	0
1	2	700	3	0
2	3	800	4	0
3	4	5000	15	1
4	5	6000	18	1
5	6	7000	20	1

```

>>> |

```

Ln: 27 Col: 0